Logistic Regression Model – Explained

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The problem

A linear model outputs any number, but a yes/no prediction needs a probability between 0 and 1.

Linear regression gives $\hat{y} = w_0 + w_1 x$, which ranges from $-\infty$ to $+\infty$. For pass/fail or defect/no-defect, a raw output of 7.3 or −2 cannot be a probability.

The fix: pass the line through a sigmoid

The sigmoid function maps any input into the range 0 to 1.

```
f(x) = \frac{1}{1 + e^{-x}}
```

- a very negative input gives about 0
- an input of 0 gives exactly 0.5
- a very positive input gives about 1

What the red arrows mean

The slide uses the letter x for two different things: put the whole linear expression into the sigmoid's placeholder slot.

- The x in $f(x) = 1/(1+e^{-x})$ is only a placeholder for "whatever goes in".
- The x in $\hat{y} = w_0 + w_1 x$ is your actual feature, such as torque or temperature.

Substituting $w_0 + w_1 x$ for the placeholder gives the logistic regression model:

```
p = \frac{1}{1 + e^{-(w_0 + w_1 x)}}
```

With many features the exponent just grows:

```
p = \frac{1}{1 + e^{-(w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n)}}
```

Numeric example

At a torque of 5 the model is undecided (p = 0.50); below that it leans fail, above it leans pass.

Suppose a model predicts whether a device passes a functional test from measured torque x, with $w_0 = -10$ and $w_1 = 2$, so $z = -10 + 2x$.

x (torque)	$z = w_0 + w_1x$	$p = 1/(1+e^{-z})$	Meaning
3	−4	0.018	almost surely fail
5	0	0.50	undecided
7	+4	0.982	almost surely pass

The straight line z is unchanged; the sigmoid only bends its output into an S-curve between 0 and 1.

Key takeaways

- **The output p is a probability, not a class.** Turn it into a class with a threshold, usually 0.5: $p \geq 0.5$ means class 1.
- **The decision boundary is where z = 0.** In the example that is $x = -w_0/w_1 = 5$. The model is still linear underneath, so logistic regression is a linear classifier.
- **The weights are not fitted by least squares.** They are fitted by maximizing likelihood (minimizing log-loss); the next slides likely cover this.